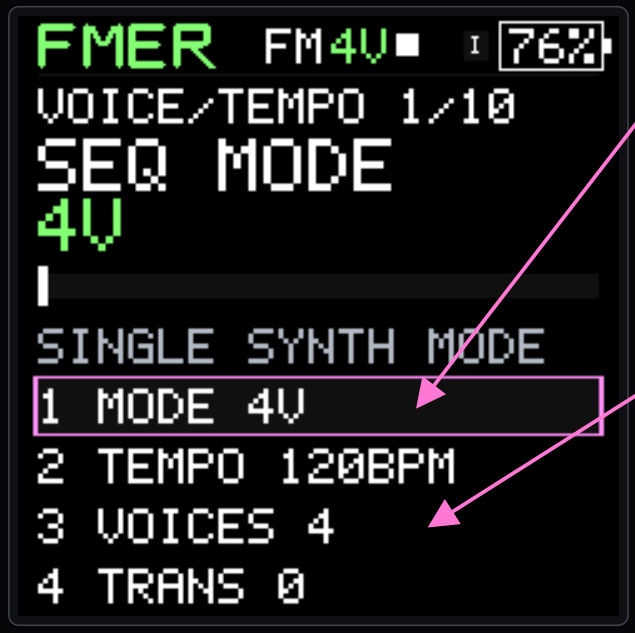
Playing modes

One instrument or four independent parts.

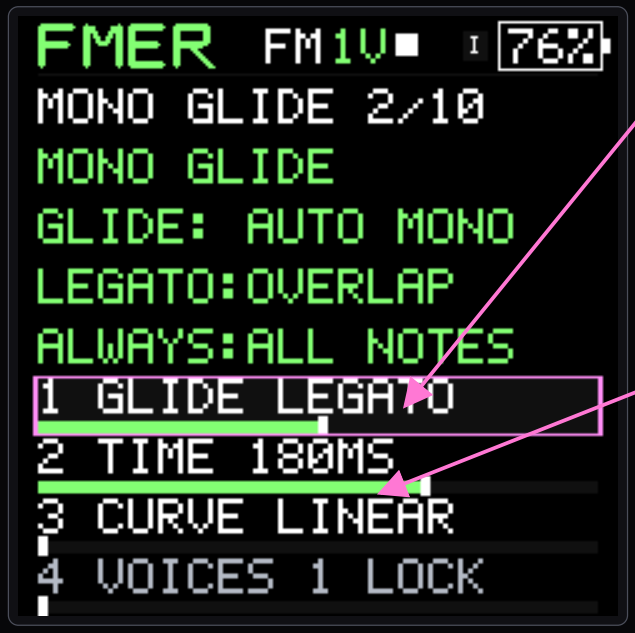
GLO / VOICE-TEMPO



01 K1 selects 1V, 4V or 4T

02 Voices follows the active allocation

GLO / MONO GLIDE



01 Choose Legato or Always

02 Set glide time and contour

01 1V: mono instrument with glide. 4V: four voices shared by one instrument.

02 4T: four mono tracks with separate instrument references.

03 Use GLO K1 or SEQ / PLAY MODE K1. Entering or leaving 4T stops playback.

Six-operator FM

Four-voice FM sound design with DX7-style voice exchange.

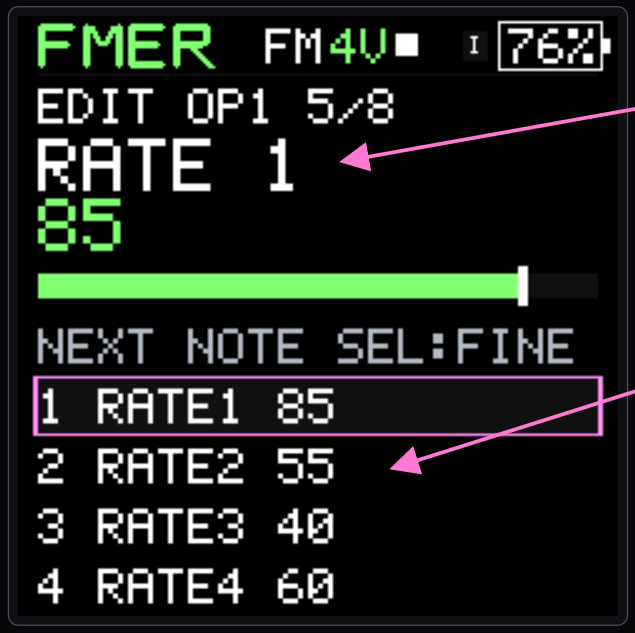
EDIT / TUNE



01 SELECT chooses the operator

02 K1–K4 edit level and frequency

EDIT / OP RATES



01 Operator envelope rates

02 Each knob edits one stage

01 Six operators, 32 algorithms, feedback and detailed rate/level envelopes.

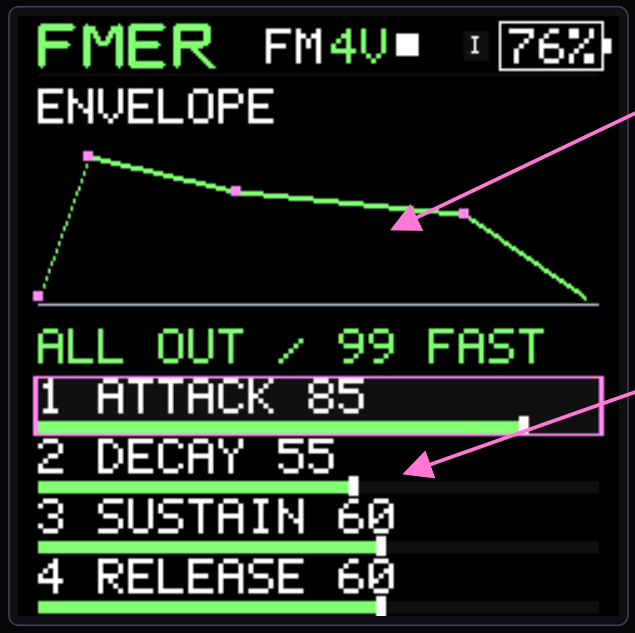
02 ALGORITHM changes the current FM sound; SAVE retains the edit.

03 Standard DX7 voice/bank import is available in Companion; exact DX7 emulation is not claimed.

FM envelopes and scaling

Shape both loudness and harmonic movement.

ENV / CARRIERS



01 Carrier envelope overview

02 Four controls edit carrier stages

EDIT / KEY SCALE



01 Operator-specific keyboard scaling

02 Breakpoint, depth and curve

01 ENV edits the carriers together; EDIT provides per-operator envelopes.

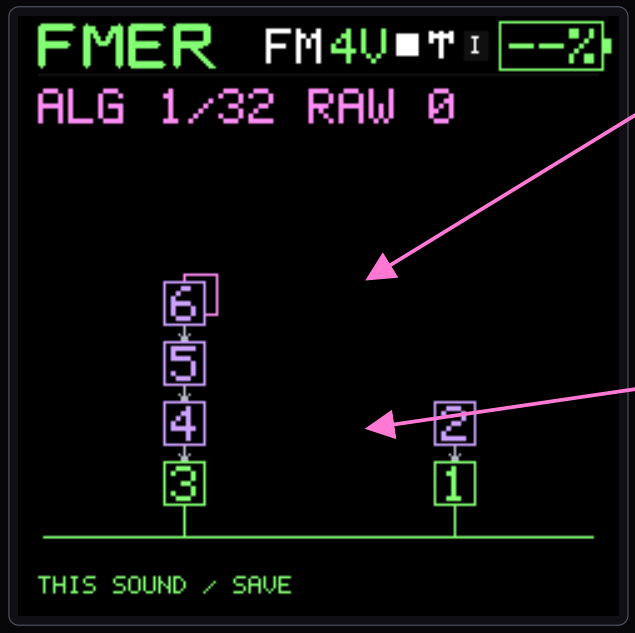
02 EDIT also includes sensitivity, key scaling and pitch-envelope pages.

03 FM rates and levels are not millisecond ADSR values.

FM algorithm diagrams

See how the six operators connect.

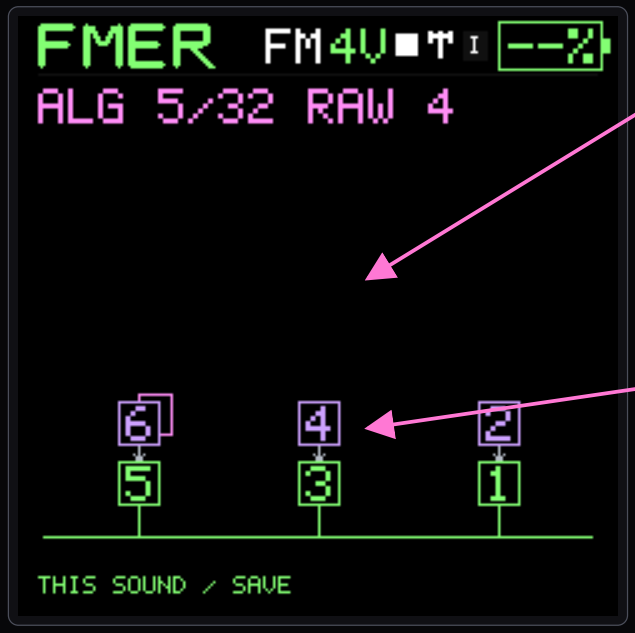
ALGORITHM / 1



01 Operator routing diagram

02 The active sound receives the edit

ALGORITHM / 5



01 Different carrier/modulator structure

02 Save the instrument to retain it

01 Turn ALGORITHM on an FM instrument to view its routing diagram.

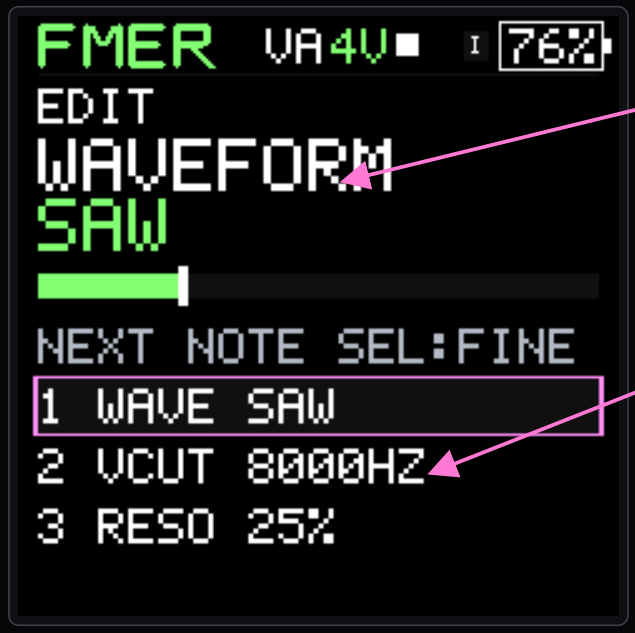
02 Carriers reach the output; modulators change another operator’s tone.

03 All 32 routing diagrams are included in the FM handbook.

Virtual Analog

A second engine for direct oscillator, filter and envelope work.

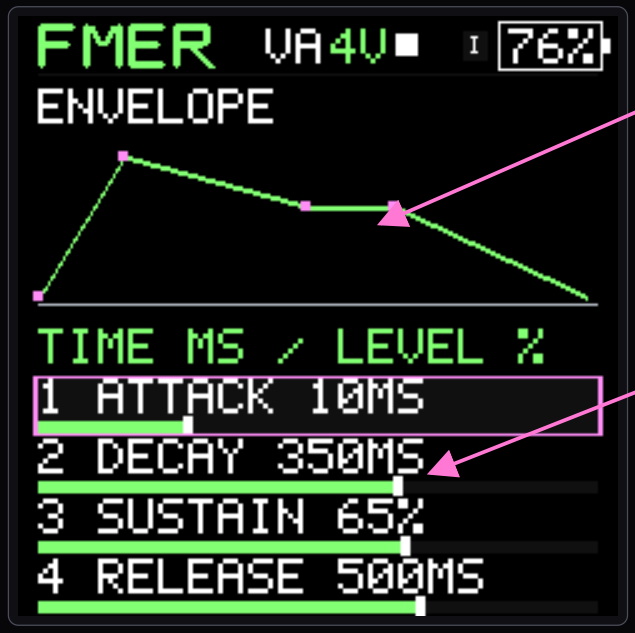
EDIT / VA



01 Waveform and voice filter

02 Cutoff and resonance are separate

ENV / VA



01 Envelope shape

02 Attack, decay, sustain and release

01 Choose sine, saw, square, triangle or noise.

02 The voice filter is distinct from the reorderable shared FX filter.

03 Use ENV for amplitude shaping and LFO for stored modulation.

128 onboard instrument slots

64 editable DEFAULT slots and 64 independent USER slots.

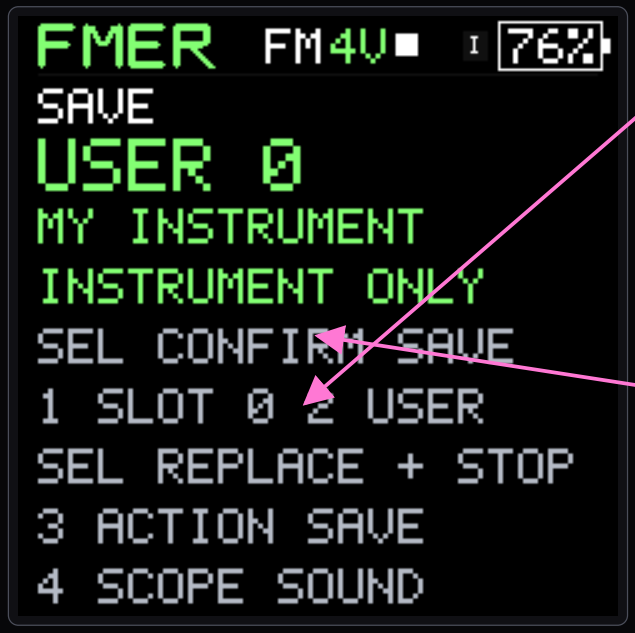
PRESETS



01 Bank and sound selection

02 PRESETS crosses all banks

SAVE / INSTRUMENT



01 Choose bank and destination

02 SEL confirms Save or Load

01 The supplied DEFAULT bank contains 32 FM and 32 VA sounds.

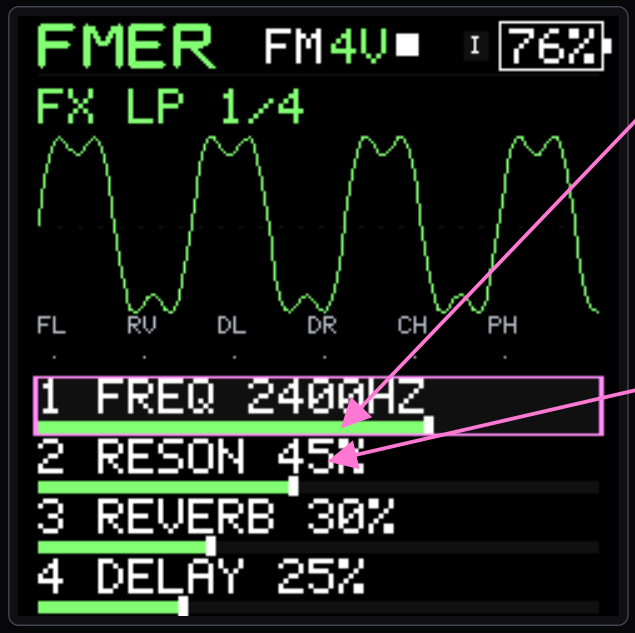
02 Either engine can occupy either bank; USER starts with independent default copies.

03 Reset to FMER Default affects DEFAULT only. Reset User Bank and Clear User Bank are separate.

Filter and resonance

Low-pass, high-pass and band-pass with directional shading.

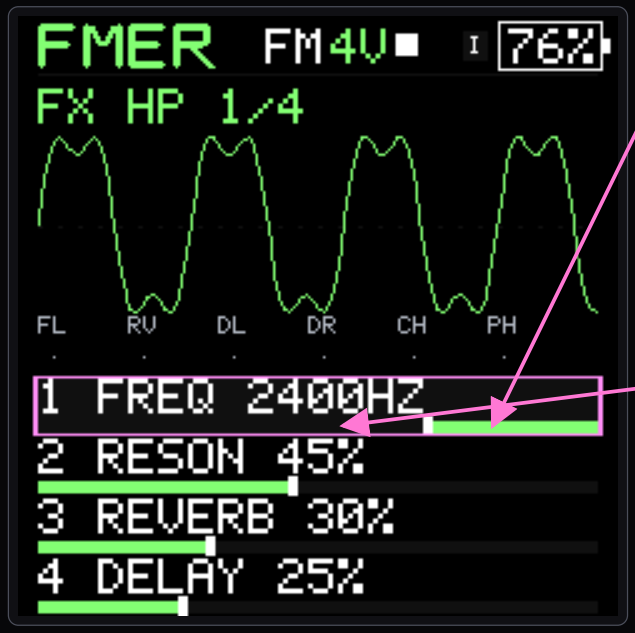
FX / LOW PASS



01 Low frequencies pass below cutoff

02 Frequency and resonance

FX / HIGH PASS



01 High frequencies pass above cutoff

02 The fill direction reverses

01 FX page 1: K1 cutoff, K2 resonance; ALGORITHM chooses filter type.

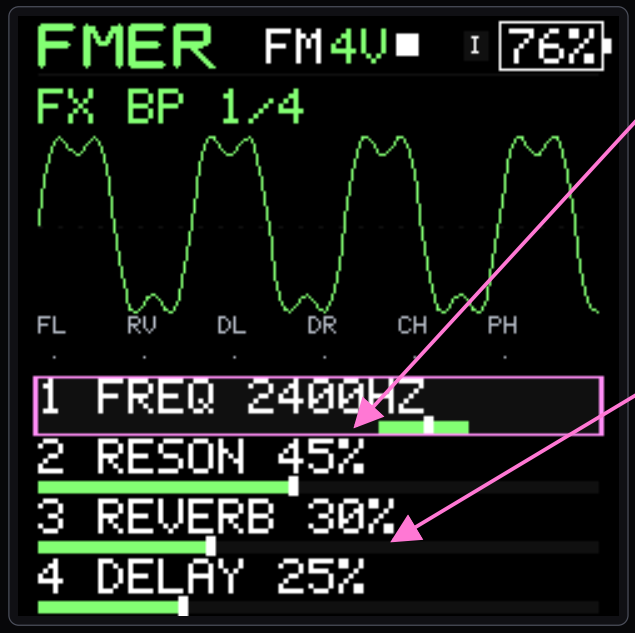
02 Frequency uses a logarithmic control scale with actual Hz visible.

03 Resonance emphasizes the region around cutoff; it is not a second frequency.

Band-pass and HOME controls

The same controls remain in the same order.

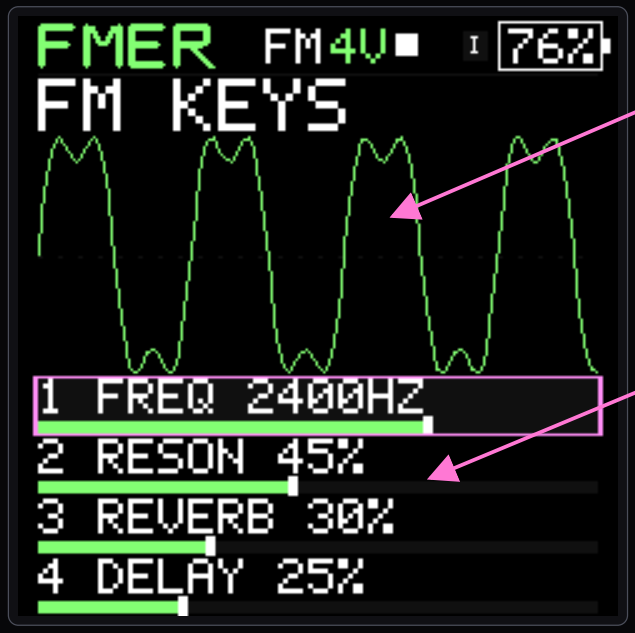
FX / BAND PASS



01 Pass band surrounds cutoff

02 K3 reverb; K4 delay

HOME / SINGLE INSTRUMENT



01 Main mix oscilloscope

02 Cutoff, resonance, reverb, delay

01 Band-pass shading surrounds the cutoff point.

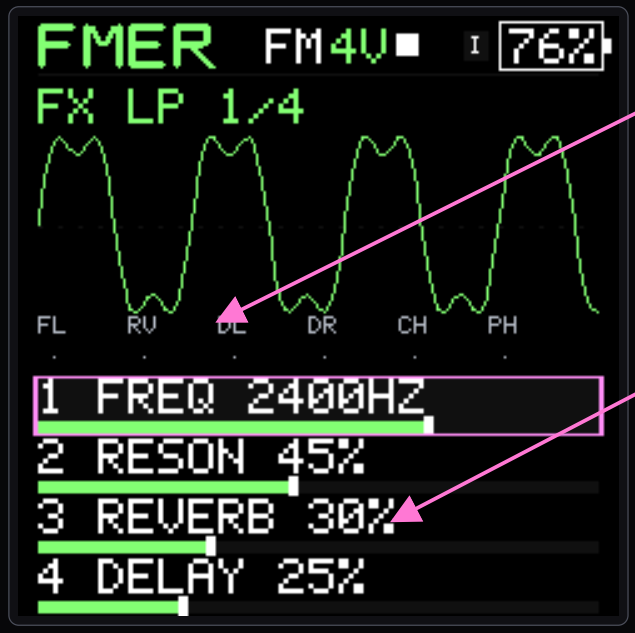
02 Single-instrument HOME mirrors the first FX page's four control positions.

03 Press FX to reach the remaining effect settings.

Four reverb models

Room, Plate, Hall and Cloud share direct return and decay controls.

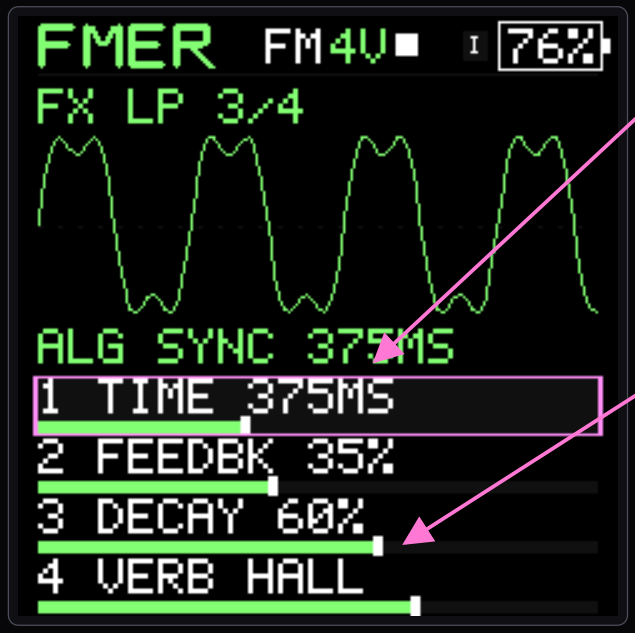
FX / FILTER + SPACE



01 Shared effect activity

02 K3 changes reverb return

FX / SPACE SETTINGS



01 Actual delay time is shown

02 K3 decay; K4 reverb model

01 Use the return amount to place the instrument in the space.

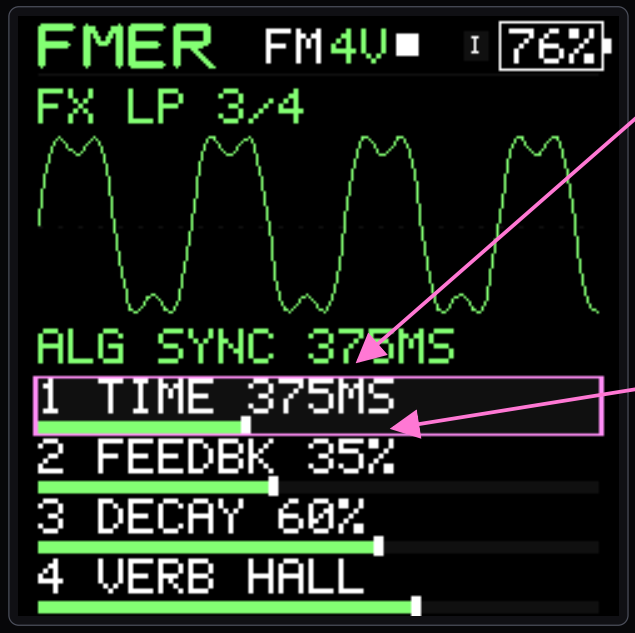
02 Select Room, Plate, Hall or Cloud on FX page 3.

03 The six-stage shared effects chain can be reordered.

Free-time or tempo-synced delay

Dial milliseconds or choose a musical division.

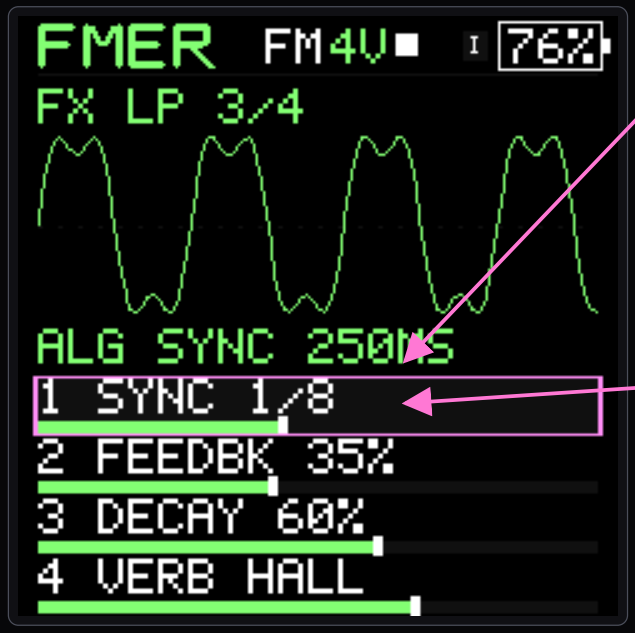
FX / FREE MS



01 Actual milliseconds

02 K1 time; K2 feedback

FX / SYNCED



01 Tempo-derived time

02 K1 division after ALGORITHM selects sync

01 Divisions: 1/4, dotted 1/8, 1/8, triplet 1/8, 1/16, triplet 1/16 and dotted 1/4.

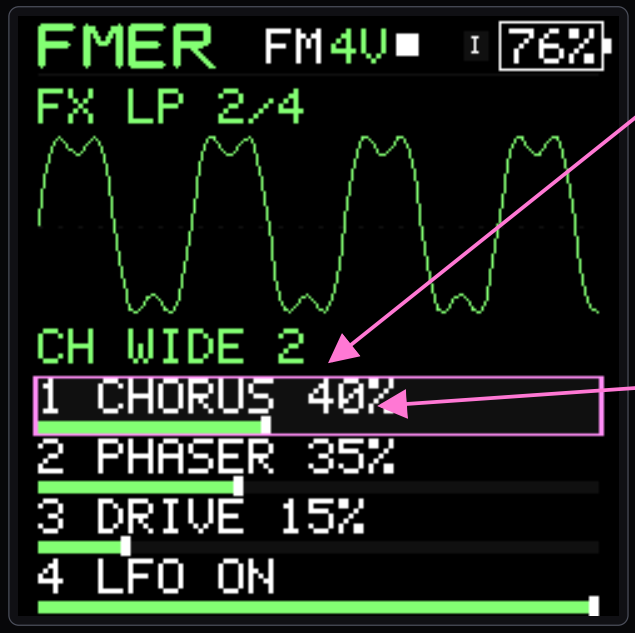
02 ALGORITHM switches Free ms / division; FX page 1 K4 controls the return.

03 Delay capacity is 20–1000 ms. MAX indicates the capacity limit.

Chorus and phaser

Four named characters for each modulation effect.

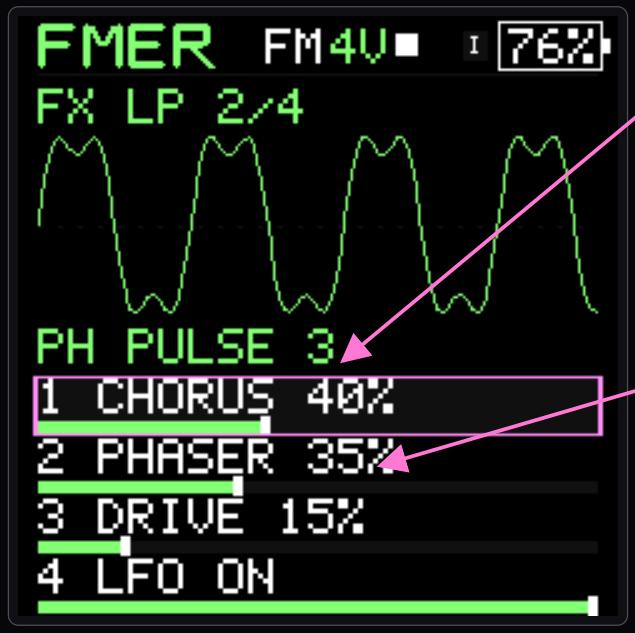
FX / CHORUS



01 Chorus character

02 K1 sets chorus amount

FX / PHASER



01 Phaser character

02 K2 sets phaser amount

01 Chorus: Dream, Wide, Shimmer, Rush. Phaser: Drift, Sweep, Pulse, Storm.

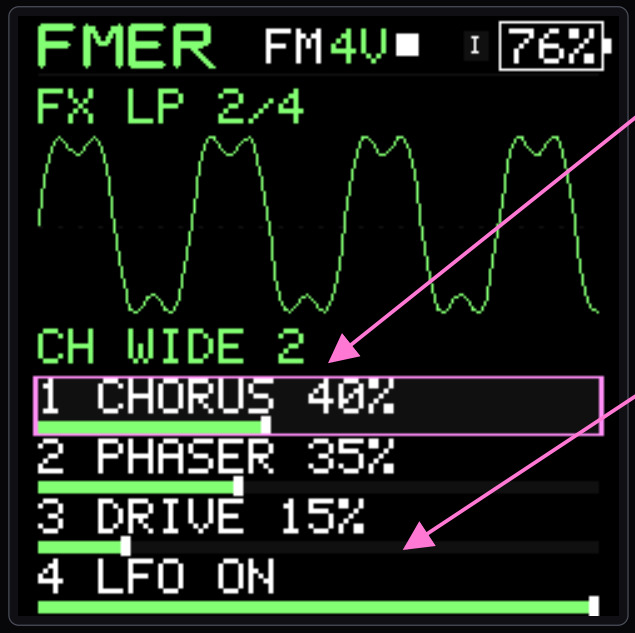
02 Turn the relevant amount knob, then ALGORITHM selects its character.

03 The implemented effect is Phaser, not Flanger.

Drive and LFO enable

Add saturation or enable the sound's stored modulation.

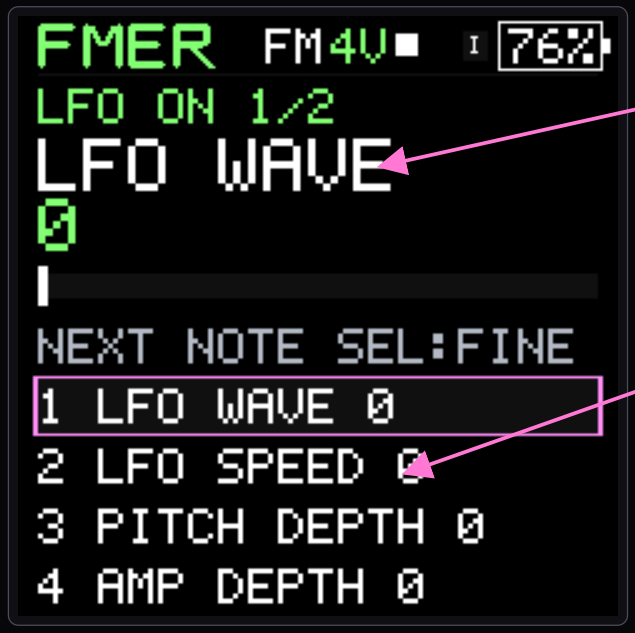
FX / MODULATION



01 Character follows the focused effect

02 K3 drive; K4 program LFO

LFO / FM MODULATION



01 FM modulation controls

02 Wave, speed, pitch and amplitude

01 Drive is the distortion stage in the reorderable audio chain.

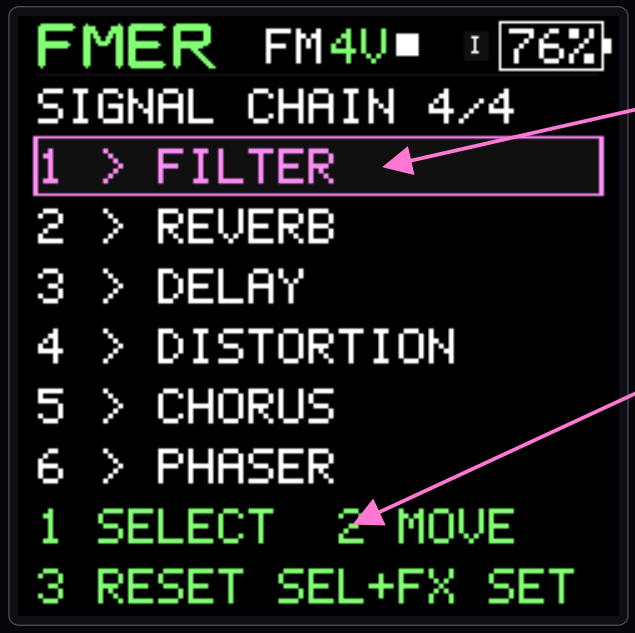
02 The LFO toggle retains its settings when disabled.

03 LFO is a modulation source, not a seventh reorderable audio effect.

Reorder the effects chain

Choose the order in which the six stages process audio.

FX / SIGNAL CHAIN



01
Selected stage and current position

02
K1 selects; K2 moves the stage

01 Filter, Reverb, Delay, Distortion, Chorus and Phaser can be reordered.

02 K3 recalls the personal default. Hold SEL, then press FX to save a new default; playback stops.

03 GLO / RESET restores original routing separately from instrument-bank resets.

Engine-specific modulation

Stored motion settings for FM and VA instruments.

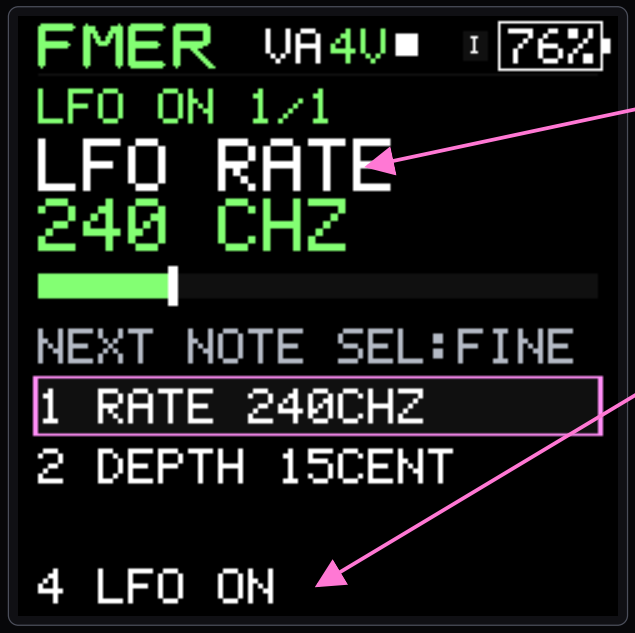
LFO / FM SYNC



01 FM-specific delay and sync fields

02 SELECT moves between LFO pages

LFO / VA



01 VA rate and depth

02 Program LFO enable

01 FM includes waveform, speed, pitch/amplitude depth and sensitivity.

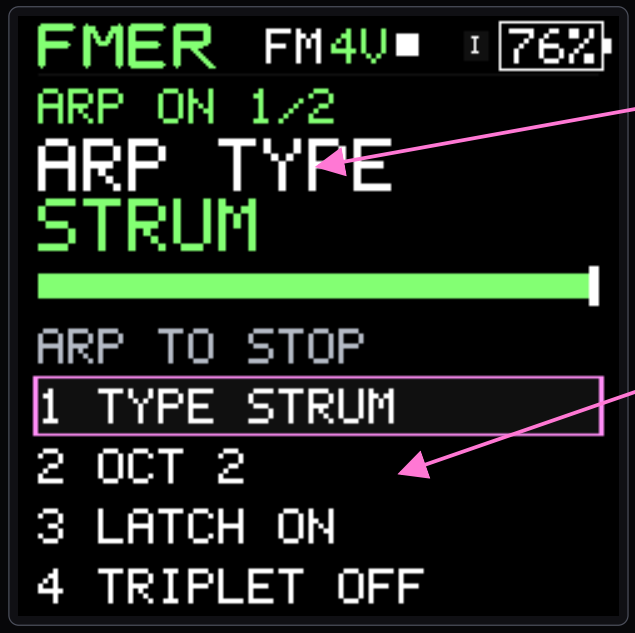
02 VA provides modulation rate and depth. FX page 2 K4 gates the stored LFO.

03 Some DX7 sync behavior differs: inspect the parameter reference for implementation limits.

Eleven arpeggiator types

Turn held notes into a timed musical part.

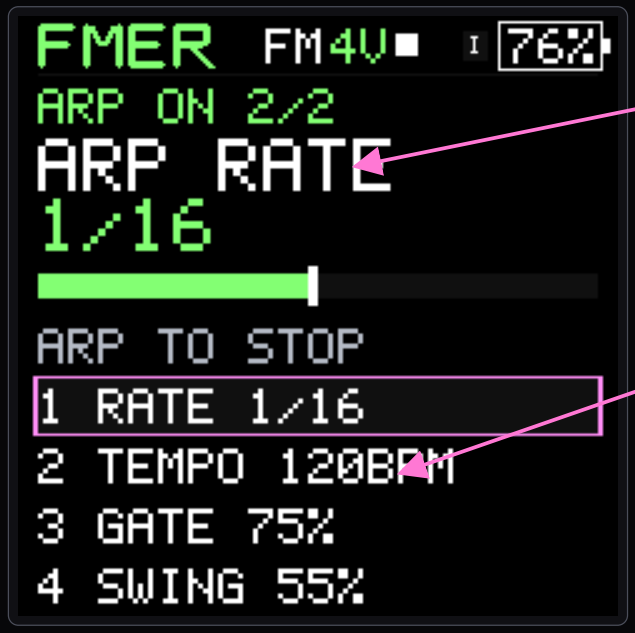
ARP / TYPE + LATCH



01 Strum is one of eleven types

02 Type, octaves, latch, triplets

ARP / TIMING



01 Timing controls

02 Division, tempo, gate and swing

01 Up, Down, Exclusive, Played, Inclusive, Random, Repeat, Down Exclusive, Down Inclusive, Outside In, Strum.

02 1–4 octaves; divisions 1/1 through 1/32, with triplets; gate and 50–75% swing.

03 Press ARP again to toggle. SELECT switches type and timing pages.

Record notes or generated arps

A four-beat count-in leads into quantized recording.

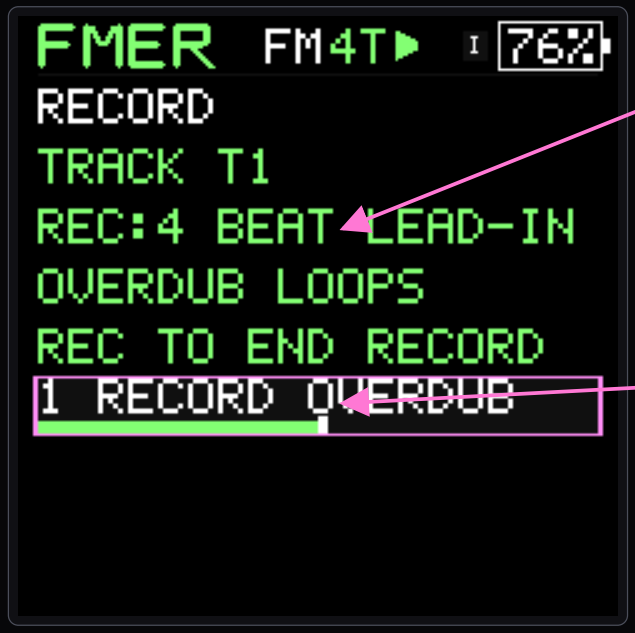
SEQ / COUNT-IN



01 Shared four-beat lead-in

02 Record mode remains visible

SEQ / RECORD



01 Choose the recording method

02 Step, Overdub or Replace

01 Overdub adds to the current pattern; Replace records a replacement pass.

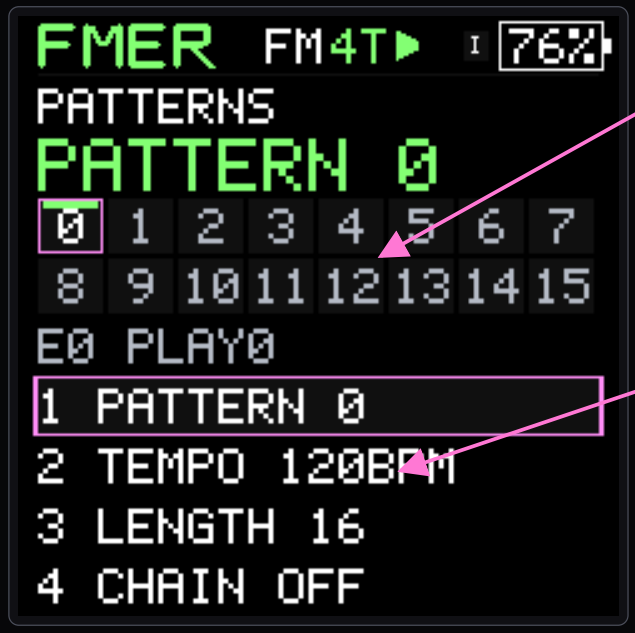
02 Live recording captures rests and held notes.

03 REC from the ARP page records generated arp notes into the selected pattern.

Sixteen patterns per project

Build phrases from 1 to 64 steps and switch them in performance.

SEQ / PATTERNS



01 Pattern bank and playback state

02 Pattern, tempo, length and chain

SIXTEEN-PATTERN OVERVIEW



01 Selection and playback markers

02 Pattern destinations are 0–15

01 Patterns are shared by the project's tracks; they are not 16 separate banks per track.

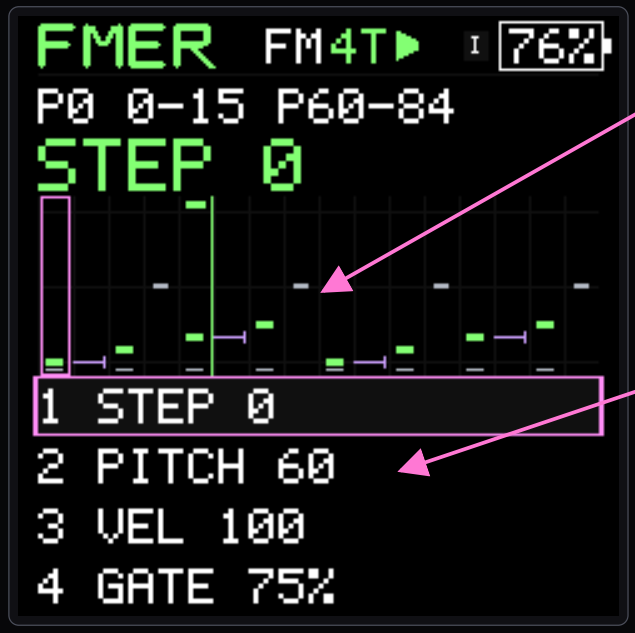
02 Each pattern can contain up to 64 steps.

03 Four-track parts are mono; single-instrument playback shares its voice pool with live keys and arp.

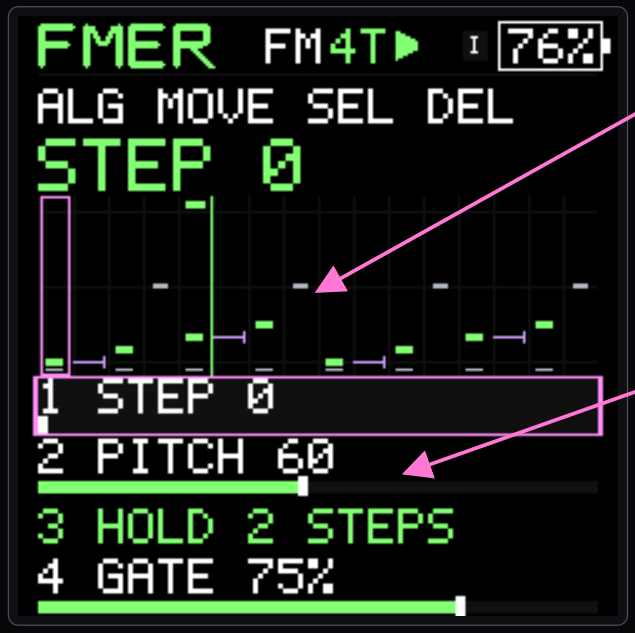
Edit notes, rests and ties

Refine a phrase without recording it again.

SEQ / STEP EDIT



SEQ / NOTE EDIT



01 NOTE EDIT: ALGORITHM moves the logical note in time; SEL deletes it.

02 Hold SEL while turning pitch down/up to insert Rest/Tie.

03 STEP EDIT changes individual steps; NOTE EDIT works with held notes.

Copy sections and chain patterns

Reuse musical material and keep it moving.

SEQ / COPY + EDIT



01 Choose source range and destination

02 SEL applies the selected action

SEQ / CHAIN MEMBERS



01 Pattern membership

02 Choose the cursor and inclusion

01 Copy, Paste, Cut, Clear and Undo operate on bounded step ranges.

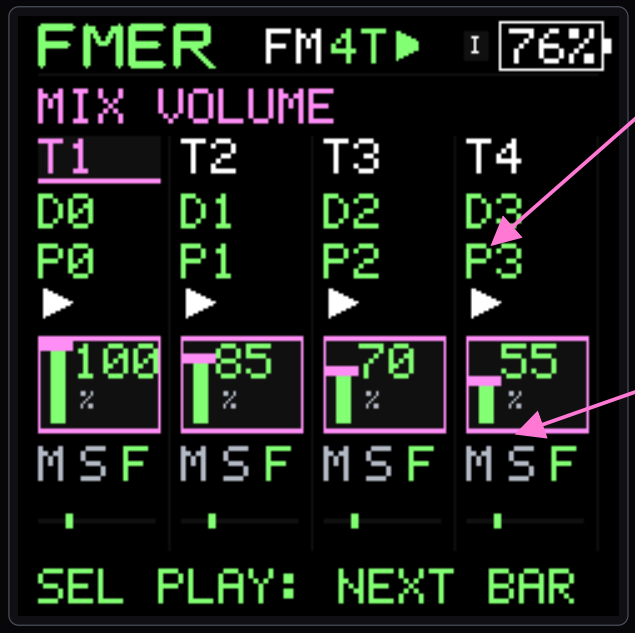
02 Pattern chaining cycles enabled members; Song mode adds ordered repeats and transpose.

03 At least one chain member remains enabled.

Four-track performance mixer

One knob per track, with a clearly selected editing layer.

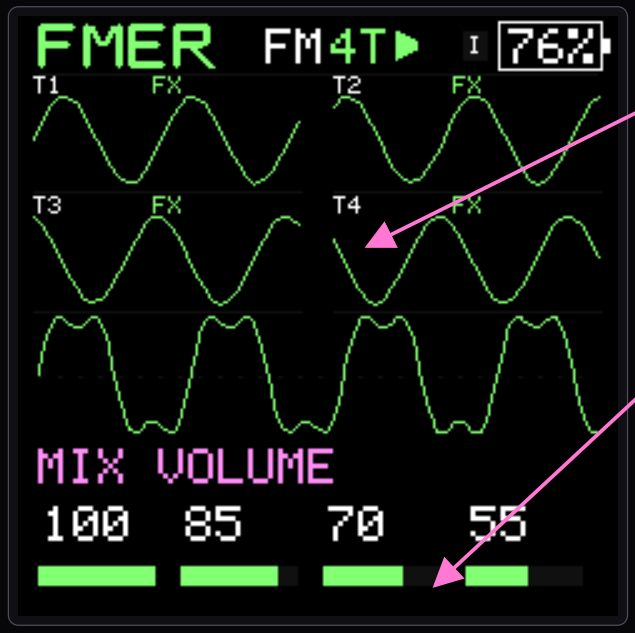
SEQ / MIXER



01 Four track strips

02 Volume, mute, solo and FX state

HOME / FOUR TRACKS



01 Part traces plus the final mix

02 Knobs 1–4 follow tracks 1–4

01 Mixer layers: Patch, Pattern, Run, Volume, Mute, Solo, FX. SELECT / ALGORITHM / SEL cycles the layer.

02 Run left stops a track; right queues its next-bar launch.

03 FX toggles each track's route through the shared chain; it is not four independent FX processors.

Song arrangements

Arrange patterns, transpositions and repeats on a shared timeline.

SEQ / SONG ENTRIES



01 Repeat blocks align to musical time

02 Entry, pattern, transpose, repeats

SEQ / SONG SETUP



01 Lane and current entry

02 Live/Song, end behavior, insert/delete

01 32 entries per lane; patterns 0–15 or a 16-step rest. Repeats 1–32; transpose −24..+24 semitones.

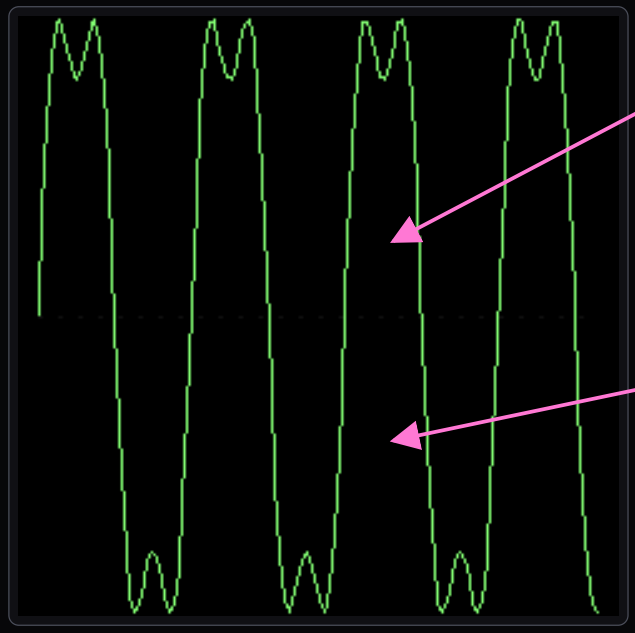
02 ALGORITHM chooses the lane; SEL switches Entries/Setup. End behavior is Stop or Loop.

03 Single-instrument modes use one active lane; 4T uses four. Save the project to retain the arrangement.

Live visual feedback

Expand the display when the sound is the focus.

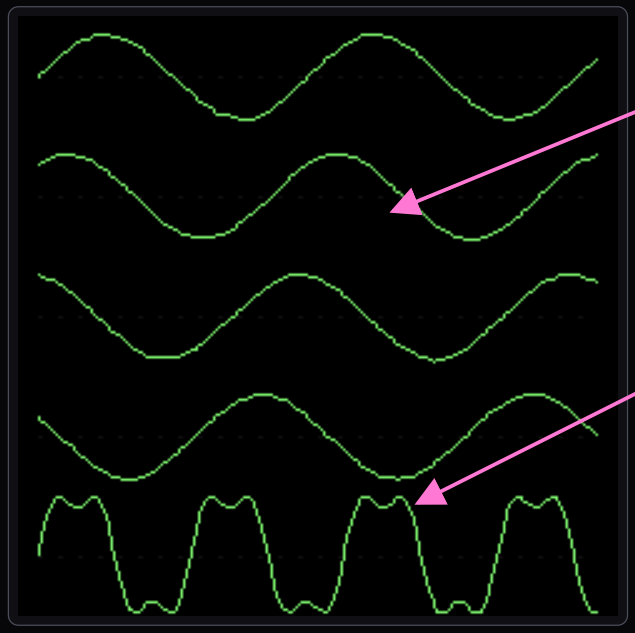
HOME / FULLSCREEN



01 Final mix waveform

02 HOME again toggles the view

HOME / FOUR-TRACK FULLSCREEN



01 Four part traces and final mix

02 Another page button exits fullscreen

01 HOME again hides the ordinary controls in either playing layout.

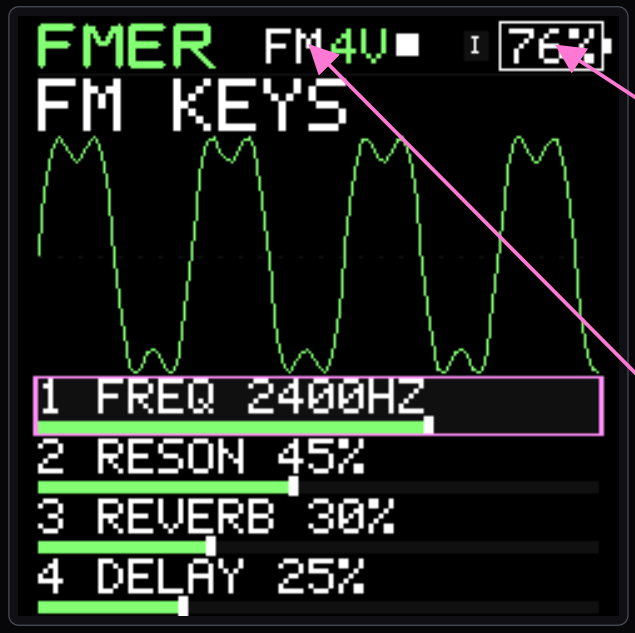
02 Part traces monitor sound before track faders and FX; the master trace shows the final mix.

03 These images use illustrative waveform data through the production renderer.

Estimated battery percentage

A numeric estimate in the compact status header.

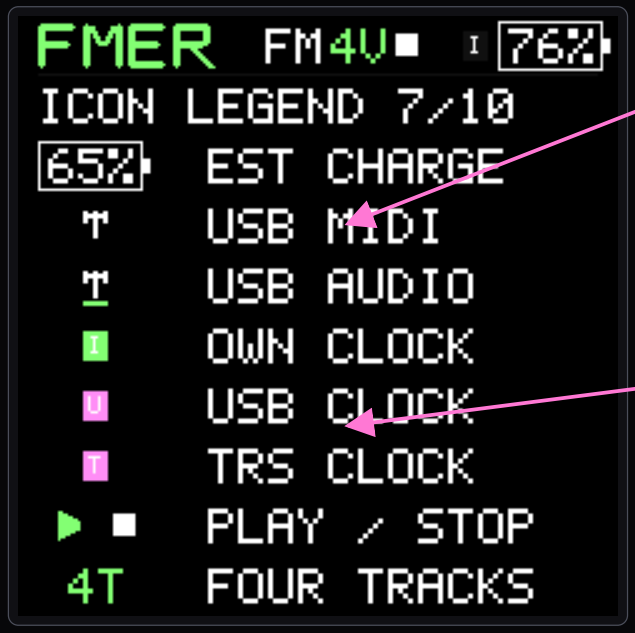
HOME / HEADER



01 Battery percentage at top right

02 Engine and playing-mode indicators

GLO / ICON LEGEND



01 Header symbols and meanings

02 Color adds status information

01 White: normal. Pink: estimate below 10%. Green: active USB data connection.

02 Green does not measure charger current. Percentage remains a calibrated estimate.

03 Save before low power; do not treat the percentage as a guaranteed remaining runtime.

Onboard projects and saving

Keep a complete working arrangement on the instrument.

SAVE / PROJECT



SAVE / INSTRUMENT



01 16 project slots retain patterns, song lanes, mix, working sound and instrument references.

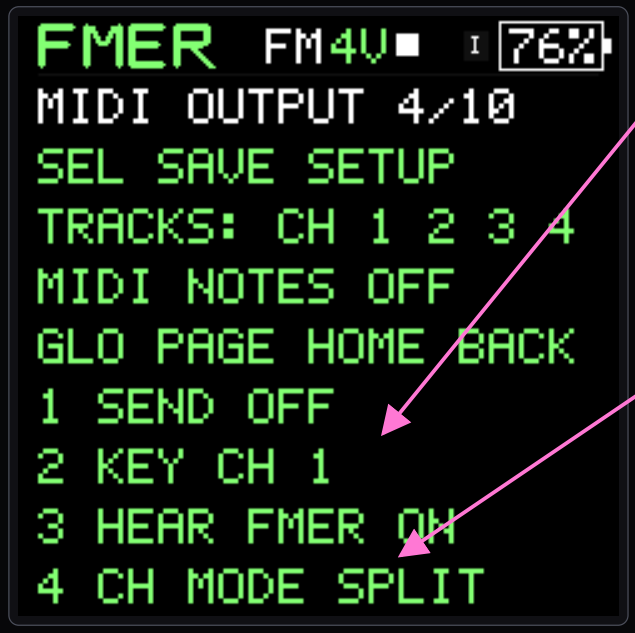
02 Track references point to bank slots; projects do not embed four independent bank instruments.

03 Save the instruments, save the project, then capture a complete unit backup.

Wired MIDI and clock

Play, synchronize and control FMER from other instruments or a DAW.

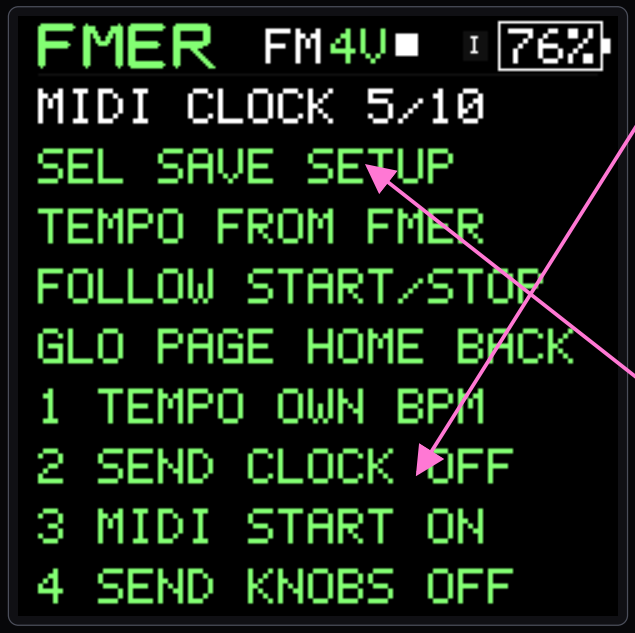
GLO / MIDI OUTPUT



01 Output route and channel

02 Local audio and Split/Merged

GLO / MIDI CLOCK



01 Clock source and transmission

02 SEL saves global MIDI settings

01 USB MIDI input/output; TRS MIDI input. Bank Select + Program Change select patches.

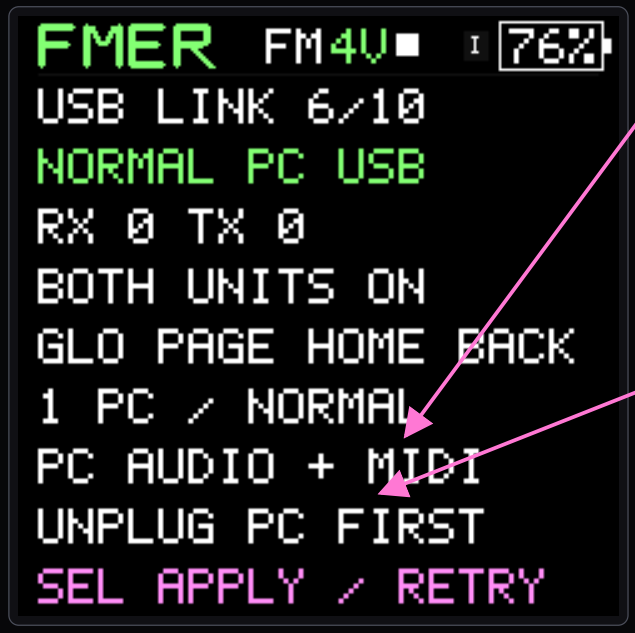
02 Split or merge generated track channels; fixed CCs and exact USB SysEx are documented in Help.

03 TRS output and Bluetooth are unavailable. USB audio is a mixed output, not four separate stems.

Direct USB instrument link

Connect two powered FMER units without a computer.

GLO / USB LINK



01
Choose PC, Master or Receiver

02
Master chooses Clock, Notes or Both

01 Unplug PC cables before applying a direct-link role; connect the units with a USB data cable.

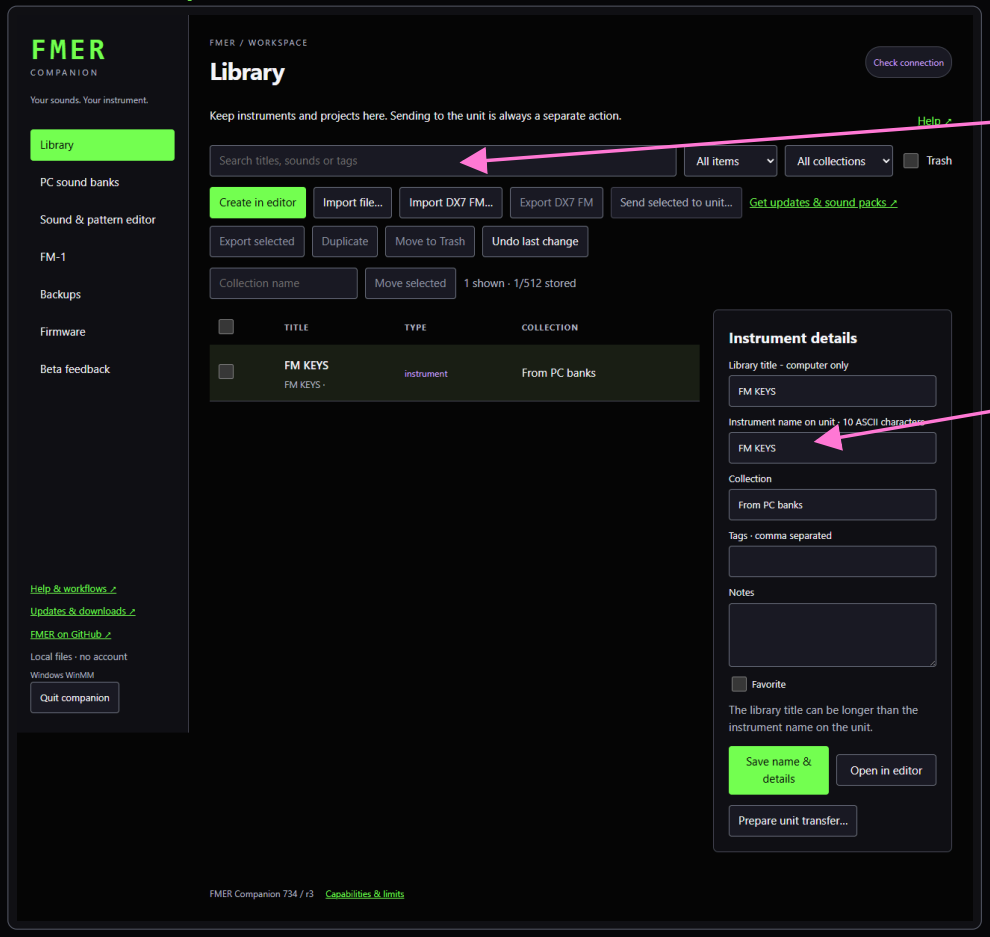
02 SEL applies. Each unit stays independently powered. One direct peer is supported.

03 Return to PC/NORMAL and press SEL before reconnecting Companion. Link roles reset at power-on.

Companion library

Organize instruments and projects locally on your computer.

COMPANION / LIBRARY



01
Search and organize the catalog

02
Name shown on the instrument

01 Collections, tags, favorites, rename, duplicate, move and Trash/Undo.

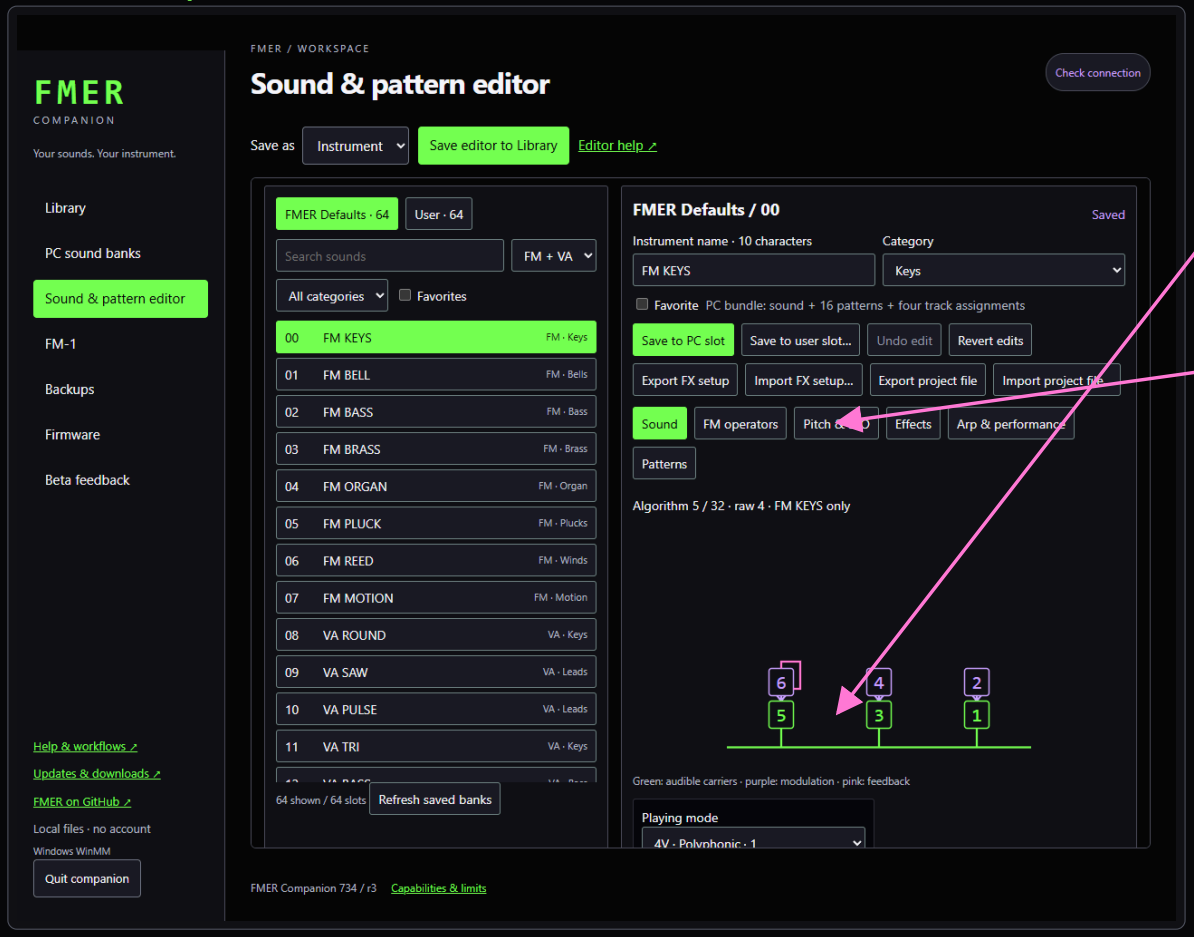
02 PC library items are separate from the unit's onboard banks and projects.

03 Import/export native FMER data and standard DX7 FM voices or 32-voice banks.

Companion editing and banks

Edit visually, audition locally and choose explicit unit destinations.

COMPANION / EDITOR



01
FM algorithm and working instrument

02
Sound, effects and sequencing tabs

01 Sliders, musical selectors and a gesture-level Undo support detailed editing.

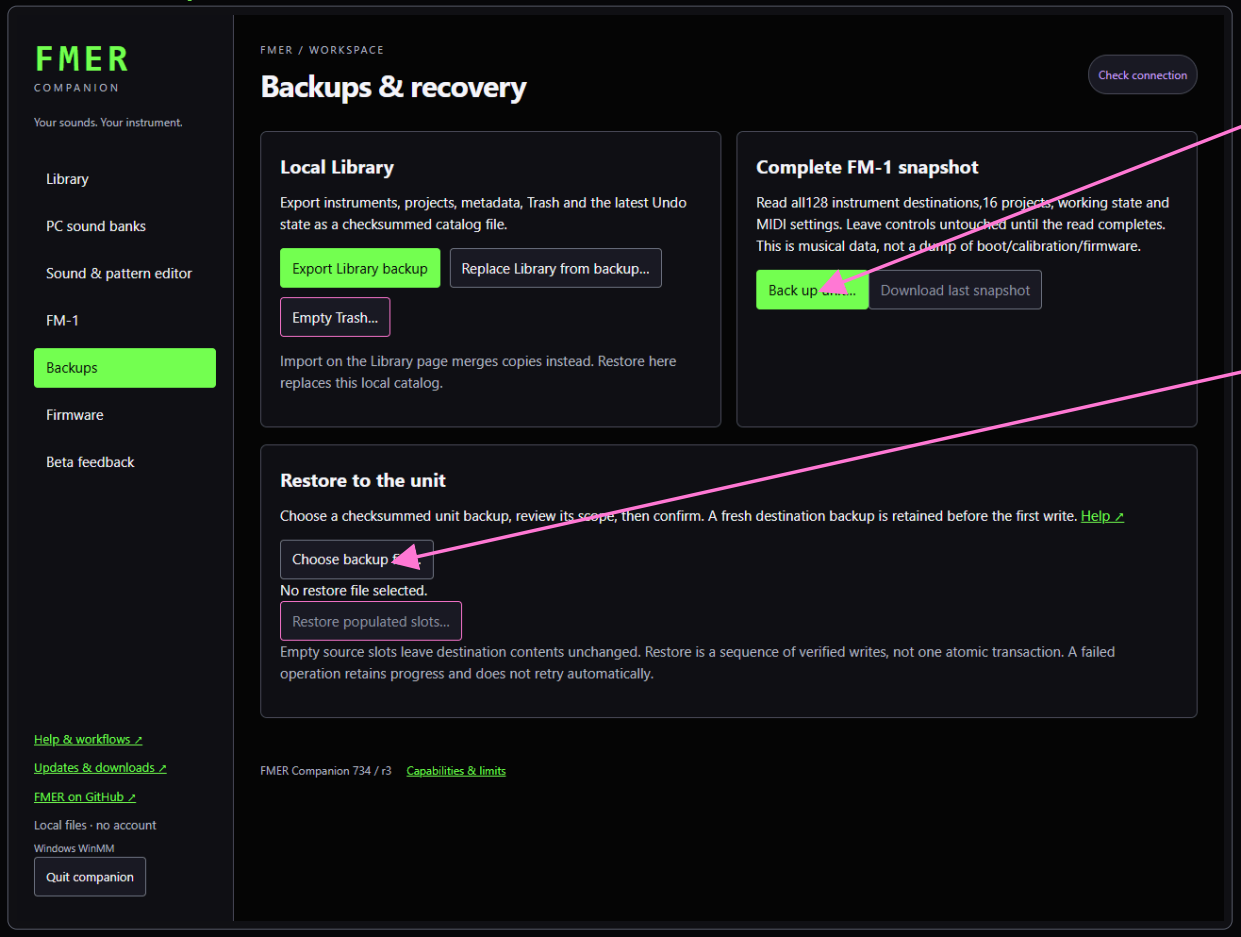
02 64 DEFAULT and 64 USER PC slots are separate from hardware storage until transferred.

03 Opening a unit instrument preserves the editor arrangement; loading a project replaces it.

Backup and restore

Keep a recoverable copy before changing an installation.

COMPANION / BACKUPS



01
Library export and unit backup

02
Choose the explicit restore scope

01 Complete supported-FMER backups include musical banks, projects, working state, MIDI and personal FX default.

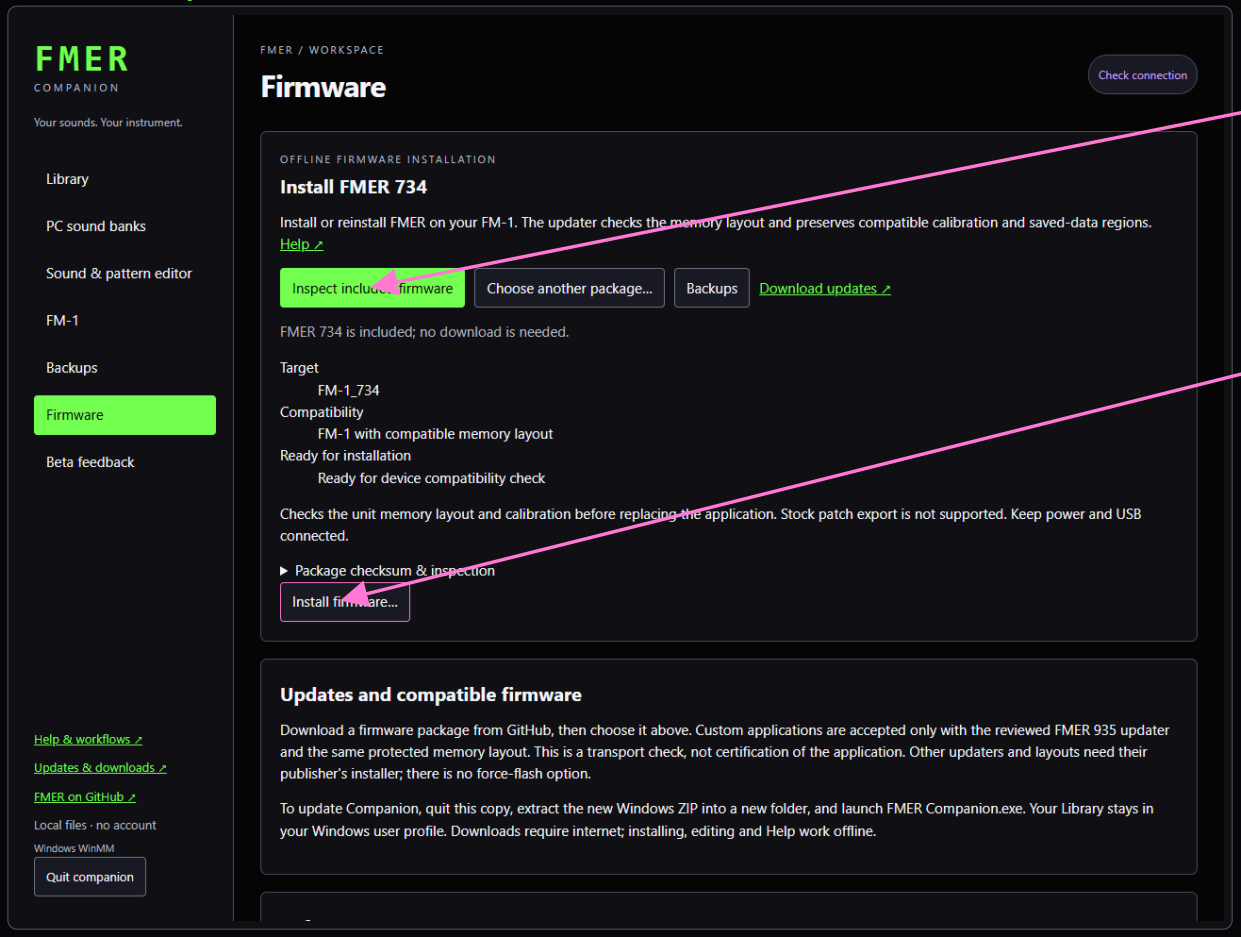
02 Keep dated archives per named instrument. Manage one connected unit at a time.

03 Restore is a verified overlay: empty source slots do not silently erase destination slots.

Offline firmware installation

Inspect, install and verify from one local Companion window.

COMPANION / FIRMWARE



01
Inspect the selected firmware

02
Install only after compatibility checks

01 Portable Windows 10/11 x64 EXE; no Python installation or hosted service required.

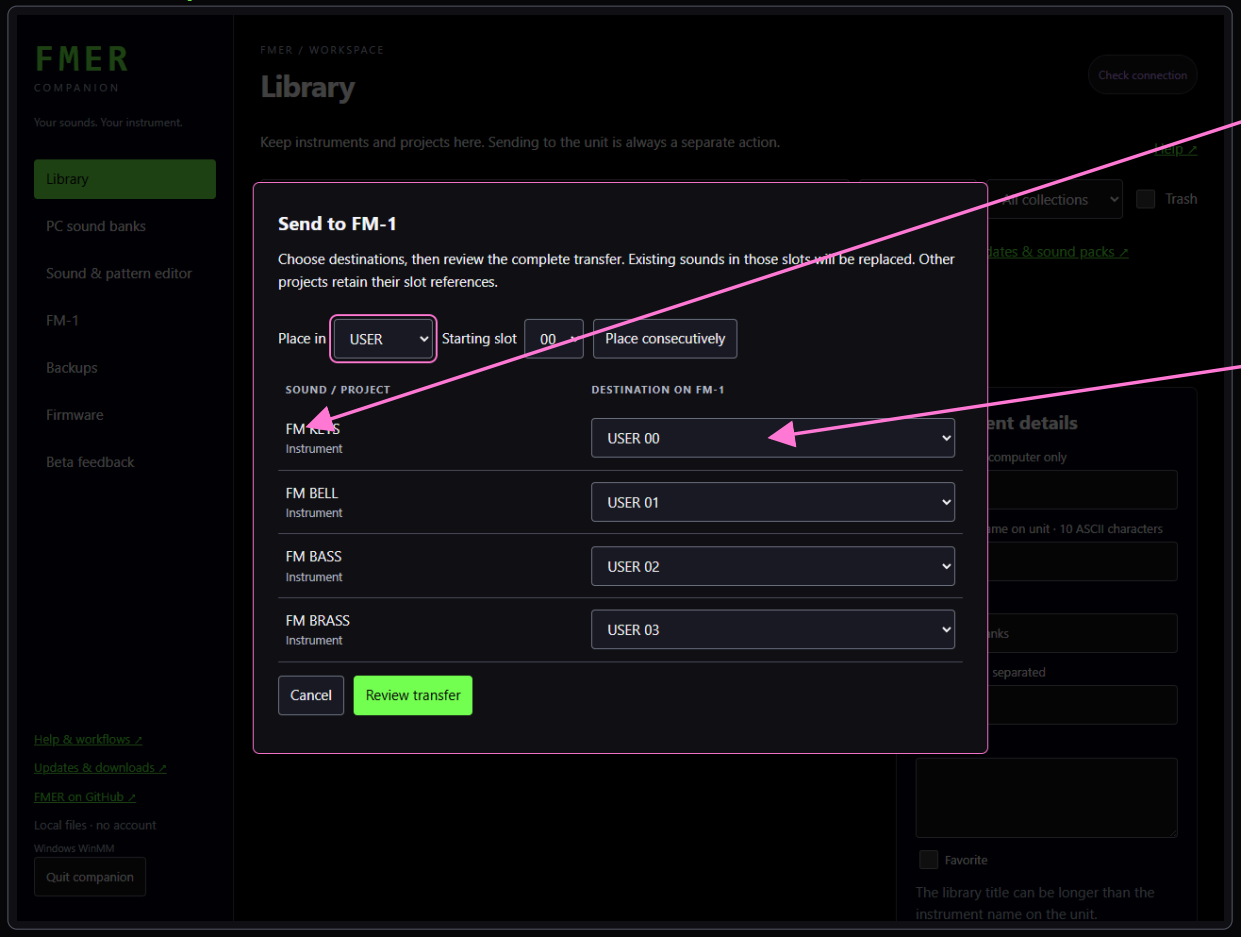
02 Use bundled or local firmware files. The app does not automatically download updates.

03 Supported layouts are checked; progress and exportable logs remain visible. Unknown layouts are not force-flashed.

Deploy sounds and projects

Review every destination before sending your set to the instrument.

COMPANION / TRANSFER REVIEW



01 Saved instrument names

02 Choose explicit unit destinations

01 Send selected Library items, a PC bank or a project with embedded sounds.

02 A complete unit backup precedes all writes; every written slot is verified.

03 Playback stops. Working state is restored afterward; a failed batch retains its completed writes.

From sound to arrangement

WORKFLOW / follow the numbered steps or read each storage layer in order

01 / CHOOSE

Select 1V,4V or 4T in GLO. Browse with PRESETS.

02 / DESIGN

EDIT, ENV, LFO and FX shape the working instrument.

03 / SAVE SOUND

SAVE / Instrument. Choose bank and slot; confirm SEL.

04 / RECORD

Choose pattern and length. REC uses Step, Overdub or Replace.

05 / ARRANGE

Assign track sounds and patterns. Add Song repeats and transpose.

06 / PRESERVE

Save Project. Back up the unit in Companion.

Tracks reference saved instrument slots. Save edited sounds before the project.

What is saved where?

WORKFLOW / follow the numbered steps or read each storage layer in order

INSTRUMENT BANKS

64 DEFAULT +64 USER. Each slot can hold FM or VA.

PATTERN SET

16 patterns in each project; each holds 1–64 steps.

SONG LANES

32 entries per lane refer to the project's patterns.

PROJECT

Working sound, patterns, lanes, mix and bank references.

PC LIBRARY

Named instruments/projects, collections and independent copies.

UNIT BACKUP

All populated banks/projects plus working/global settings.

A project references track instruments. Moving a bank instrument can change projects that reference its slot.

Choose the right playing mode

WORKFLOW / follow the numbered steps or read each storage layer in order

1V / MONO

One instrument and glide for connected lead or bass lines.

4V / POLY

One instrument; keys, arp and one sequence share four voices.

4T / PARTS

Four independent mono tracks, each with an instrument reference.

DIRECT ACCESS

GLO K1 or SEQ / PLAY MODE K1 changes allocation explicitly.

PERFORM

4T opens Mixer. Single-instrument modes open one-lane Song.

RETURN

Hold SEQ 0.8 s to return. Use HOME for scopes; FX for effects.

Entering/leaving 4T stops playback. Ordinary page navigation and preset browsing do not change mode.

Two-unit USB performance

WORKFLOW / follow the numbered steps or read each storage layer in order

01 / PREPARE

Keep both units powered. Disconnect their computer cables.

02 / CHOOSE ROLES

GLO / USB LINK: Master on one, Receiver on the other.

03 / APPLY + CONNECT

SEL applies. Join the two USB-C ports with a data cable.

04 / SET TRAFFIC

Master: Clock, Notes or Both. Avoid unintended note doubling.

05 / PERFORM

Leader transport controls the shared time; each keeps its own sounds.

06 / RETURN TO PC

Choose PC/NORMAL and press SEL before Companion use.

One direct peer. No direct USB audio between units, USB hub mode or Bluetooth group support.

Companion installation workflow

WORKFLOW / follow the numbered steps or read each storage layer in order

01 / CONNECT

One unit. Close other applications holding its MIDI ports.

02 / BACK UP

Capture supported-FMER banks/projects; name the archive by unit/date.

03 / INSPECT

Select bundled or local firmware. Review compatibility and scope.

04 / INSTALL

Confirm. Keep power and USB connected while progress is active.

05 / VERIFY

Wait for completion and the new firmware identity.

06 / REVIEW

Check sounds/projects. On failure, export the operation log.

Stock firmware cannot provide FMER musical backups. Unknown layouts require diagnosis, not a forced write.

Shortcuts / one-sheet reference

Hold the modifier first. SELECT is the rotary encoder; SEL is the button.

Hold SEL → press SEQ	Open Song and select four-track mode
Hold HOME → press SEQ	Open Mixer and select four-track mode
Hold SEL → press HOME	Toggle four-track / four-voice instrument
Hold ENV → press HOME	Select mono instrument with legato glide
Hold LFO → press HOME	Select four-voice instrument; glide off
Hold SEL → press FX	Save current FX order as personal default
Hold FX / SEQ / EDIT / GLO / LFO	After 0.8 s, return to that section’s landing page
Press HOME again on HOME	Toggle expanded oscilloscope view
Press ARP again / LFO again	Toggle the respective stored function
Hold SEL + turn an ordinary parameter knob	Use fine adjustment; context actions take priority
Hold SEL + press PLAY	Queue selected track for the next 16-step bar
Hold SEL + double-press PLAY	Queue all tracks for the next 16-step bar
NOTE EDIT: ALGORITHM / SEL	Move the logical note in time / delete it
NOTE EDIT: hold SEL + turn pitch down/up	Insert Rest / Tie
PRESETS / GLO K1	Browse sounds across banks / choose 1V,4V,4T
SAVE: choose scope, slot, action → SEL	Confirm instrument or project Save/Load

Mode changes involving 4T, SAVE/LOAD and saving a personal FX default stop playback. Track launch shortcuts are unavailable during recording.